

Algorithms for Data Science

Lecture 5: Algorithms for feature reduction



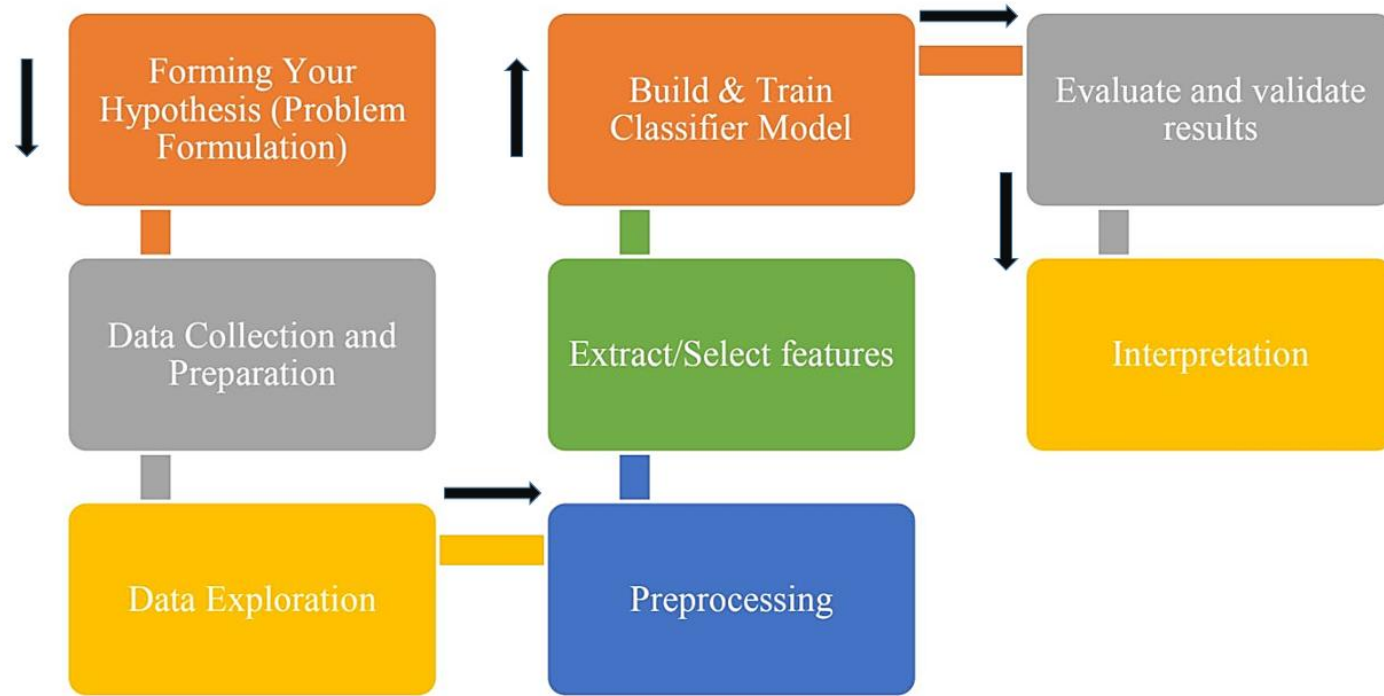


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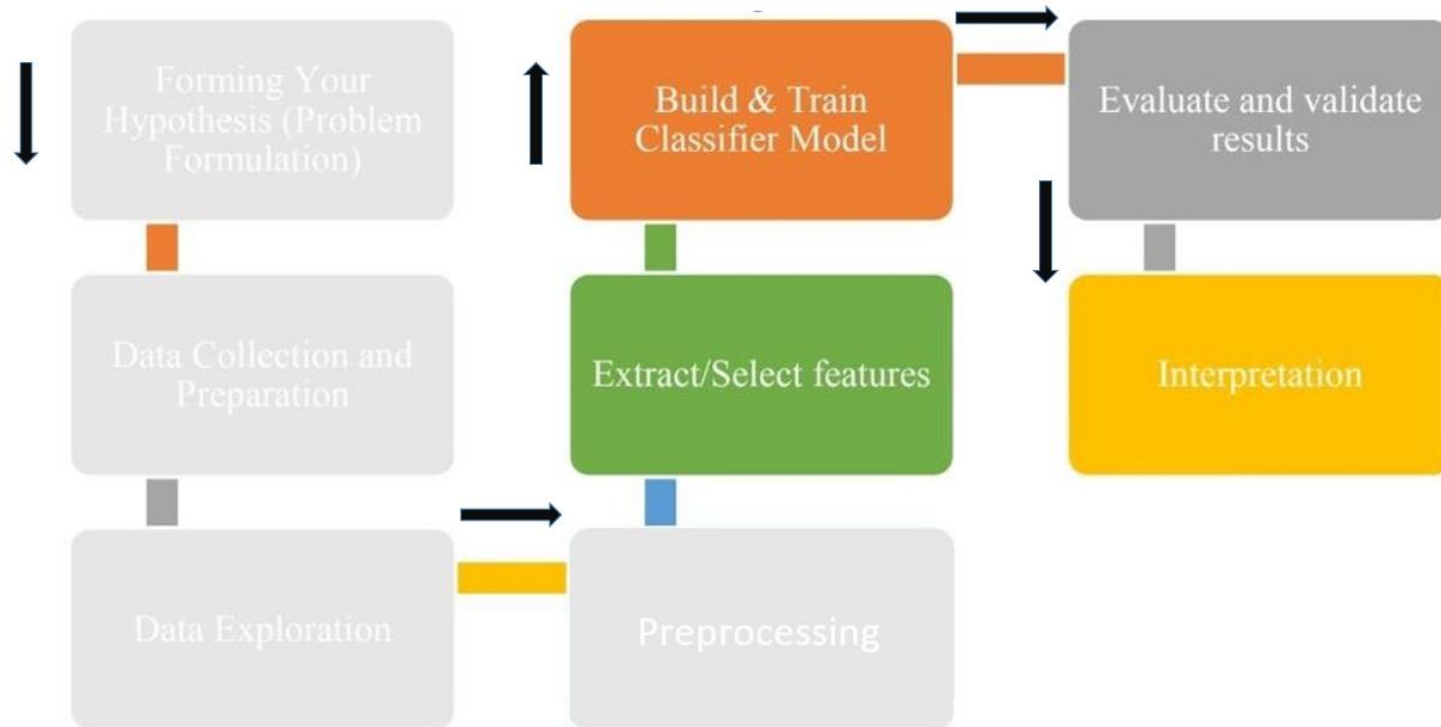
- Feature Extraction
- Principal Component Analysis (PCA)



The Data Science Pipeline



The Data Science Pipeline



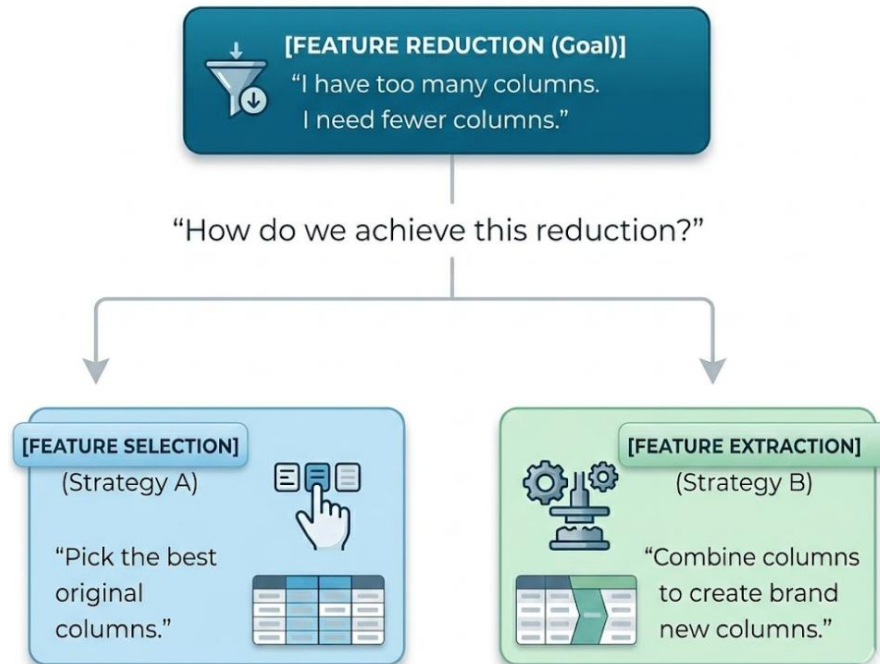
Feature Reduction

- **Feature Reduction** is the process of reducing the number of input variables (features) in a dataset.
- **Why Feature Reduction?**
 - ✓ **Reduces Dimensionality:** High-dimensional data can lead to the curse of dimensionality, where the data becomes sparse and models struggle to generalize.
 - ✓ **Improves Computational Efficiency:** Fewer features mean faster training and inference times.
 - ✓ **Handles Multicollinearity:** Feature reduction can address issues where features are highly correlated.
 - ✓ **Visualization:** Reducing features to 2 or 3 dimensions makes it easier to visualize data.



Excerpts from the previous lectures (1/2)

- Feature Reduction is the umbrella term that covers two distinct methods: Feature Selection, and Feature Extraction.



Excerpts from the previous lectures (2/2)

- **Feature Extraction:** Mathematically combining multiple columns to create new, fewer columns.
- *Example:* Combining "Height" and "Width" into a single new column called "Area," and then deleting the originals.
- **Common Feature Extraction Techniques**
 - Principal Component Analysis (PCA)
 - Linear Discriminant Analysis (LDA)
 - t-SNE (t-Distributed Stochastic Neighbor Embedding)



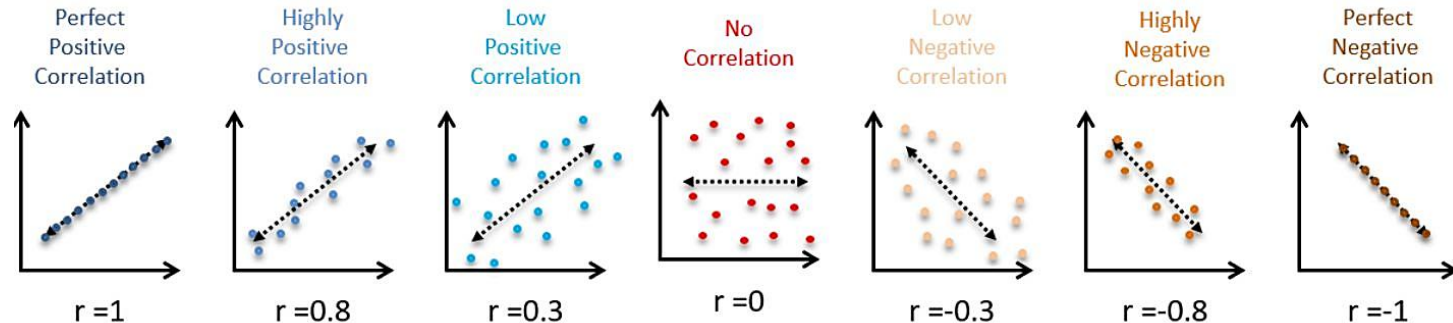
Today one of the Feature Reduction Technique will be covered: Principal Component Analysis (PCA). Let's first discuss some important terminologies.



Covariance

- The covariance is a measure of how two variables change together. It indicates the direction of the linear relationship between the variables:
 - ✓ If the covariance is positive, the variables tend to increase or decrease together.
 - ✓ If the covariance is negative, one variable tends to increase when the other decreases.
 - ✓ If the covariance is zero, there is no linear relationship between the variables.

Scatter Plots & Correlation Examples



Covariance

Mathematical Definition

For two random variables X and Y , the **covariance** is defined as:

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

where:

- μ_X = mean of X ,
- μ_Y = mean of Y ,



Covariance Matrix

- The covariance matrix is a square matrix that summarizes the variances and covariances between all pairs of features in a dataset. It is a fundamental tool in statistics and machine learning, particularly in techniques like Principal Component Analysis (PCA).

For a dataset with p features, the covariance matrix is a $p \times p$ symmetric matrix where:

- The **diagonal elements** represent the **variances** of each feature.
- The **off-diagonal elements** represent the **covariances** between pairs of features.

The covariance matrix provides a comprehensive view of the relationships between all features in the dataset.

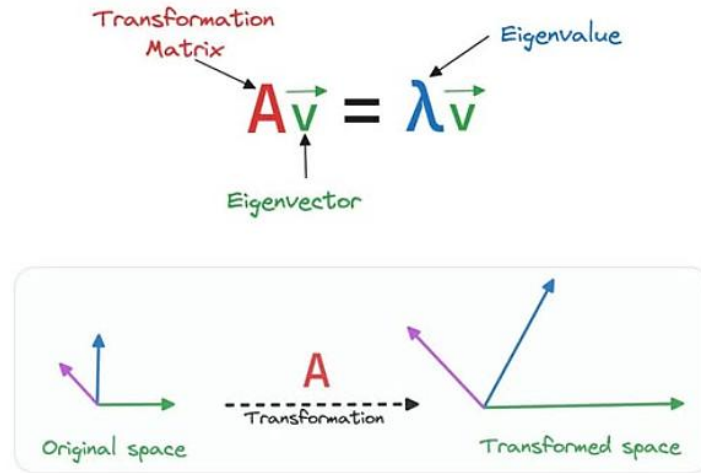
The covariance matrix Σ is:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{bmatrix}$$



Eigenvalues and eigenvectors

- In linear algebra, a linear transformation can be represented by a square matrix . When this transformation is applied to a vector, it typically changes both the vector's magnitude and its direction. However, certain special vectors maintain their direction.



The pink and green vectors are eigenvectors.

When transformed by A , they only scale (by λ) but the direction remains same.



Eigenvalues and eigenvectors

- **Eigenvector (x):** A non-zero vector x that, when multiplied by a square matrix A , results in a scalar multiple of itself. Geometrically, the vector stays on the same span (line) through the origin.
- **Eigenvalue (λ):** The scalar factor by which the eigenvector is stretched, shrunk, or reversed during the transformation.
- **Fundamental Equation:** This relationship is expressed as:

$$Ax = \lambda x$$

- **The Characteristic Equation:** To find the eigenvalues of a matrix A , we solve for λ such that the matrix $(\lambda I - A)$ is singular (non-invertible), meaning its determinant is zero:

$$\det(\lambda I - A) = 0$$

Eigenvalues and eigenvectors: Example

- Find the eigenvalues and eigenvectors for the following matrix.

Consider the matrix $A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$.

Step 1: Solving for Eigenvalues

Using $\det(\lambda I - A) = 0$:

$$\det \begin{bmatrix} \lambda - 2 & 12 \\ -1 & \lambda + 5 \end{bmatrix} = (\lambda - 2)(\lambda + 5) + 12 = \lambda^2 + 3\lambda + 2 = 0$$

Factoring $(\lambda + 1)(\lambda + 2) = 0$ gives $\lambda_1 = -1$ and $\lambda_2 = -2$.



Eigenvalues and eigenvectors: Example

Step 2: Finding the Eigenvector via Parameterization ($\lambda = -1$)

To find the eigenvector x , we solve $(-1I - A)x = 0$:

$$\begin{bmatrix} -1-2 & 12 \\ -1 & -1+5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Parameterization Process:

1. From the simplified row: $-x_1 + 4x_2 = 0 \implies x_1 = 4x_2$.
2. We identify x_2 as the free variable. Let $x_2 = t$.
3. Substituting the parameter: $x_1 = 4t$.
4. The general solution is $x = \begin{bmatrix} 4t \\ t \end{bmatrix} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.
5. Setting $t = 1$, the eigenvector is $v_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.



Eigenvalues and eigenvectors: Example

Step 3: Finding the Eigenvector via Parameterization ($\lambda = -2$)

To find the second eigenvector, we solve $(-2I - A)x = 0$:

$$\begin{bmatrix} -2 - 2 & 12 \\ -1 & -2 + 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 & 12 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

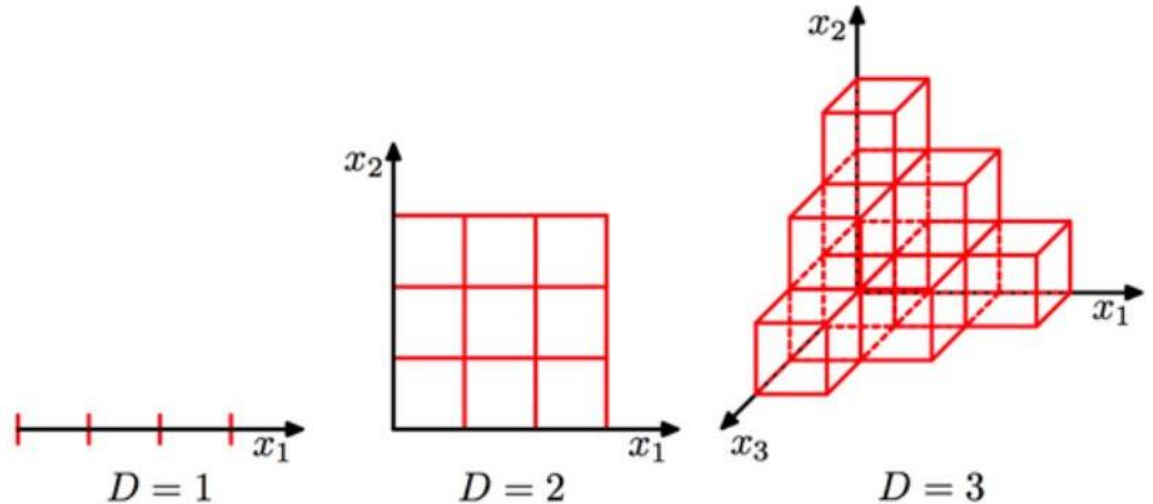
Parameterization Process:

1. From the second row:
2. We identify x_2 as the free variable. Let $x_2 = t$.
3. Substituting the parameter: $x_1 = 3t$.
4. The general solution is $x = \begin{bmatrix} 3t \\ t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.
5. Setting $t = 1$, the eigenvector is $v_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.



Principal Component Analysis (PCA)

- Principal Component Analysis (PCA) is a powerful dimensionality reduction technique used to transform high-dimensional data into a lower-dimensional space while preserving as much variance (information) as possible. It is widely used in data analysis, machine learning, and visualization.



Principal Component Analysis (PCA)- Steps

- Steps of PCA
 - Step 1: Compute the Mean of Each Feature
 - Step 2: Center the Data
 - Step 3: Compute the Population Covariance Matrix
 - Step 4: Perform Eigenvalue Decomposition
 - Step 5: Form the Principal Components



Principal Component Analysis (PCA)- Step 1

Step 1: Compute the Mean of Each Feature

The first step in PCA is to compute the mean of each feature (column) in the dataset. This is necessary to center the data.

Mathematical Formula

For each feature X_j , compute the mean:

$$\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$



Principal Component Analysis (PCA)- Step 2

Step 2: Center the Data

Centering the data means subtracting the mean of each feature from the data. This ensures that the mean of each feature is **zero**.

Mathematical Formula

For each feature X_j , center the data:

$$z_{ij} = x_{ij} - \mu_j$$



Principal Component Analysis (PCA)- Step 3

Step 3: Compute the Population Covariance Matrix

The covariance matrix captures the relationships between features. For centered data, the population covariance matrix is:

$$\text{Cov}(Z) = \frac{1}{n} Z^T Z$$

where:

- Z = centered data matrix,
- n = number of samples.



Principal Component Analysis (PCA)- Step 4

Step 4: Perform Eigenvalue Decomposition

Eigenvalue decomposition is used to find the **eigenvalues** and **eigenvectors** of the covariance matrix. These represent the principal components and the amount of variance they explain.

Mathematical Formula

Solve the characteristic equation:

$$\det(\text{Cov}(Z) - \lambda I) = 0$$

where:

- λ = eigenvalue,
- I = identity matrix.

The eigenvalues (λ) represent the variance explained by each principal component, and the eigenvectors represent the directions of the principal components.



Principal Component Analysis (PCA)- Step 5

Step 5: Form the Principal Components

The eigenvectors corresponding to the eigenvalues are the **principal components**. Normalize the eigenvectors to have a unit length.

Mathematical Formula

For each eigenvalue λ , solve:

$$(\text{Cov}(Z) - \lambda I)\mathbf{v} = 0$$

Normalize the eigenvector $\mathbf{v}$:

$$\mathbf{v}_{\text{normalized}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$



PCA- Example

Dataset

Consider the following dataset with 2 features (X_1 and X_2) and 3 samples:

Sample	X_1	X_2
1	1	2
2	2	3
3	3	4



PCA- Example

Step 1: Compute the Mean of Each Feature

For the population, the mean of each feature is calculated as:

$$\mu_{X_1} = \frac{1 + 2 + 3}{3} = 2, \quad \mu_{X_2} = \frac{2 + 3 + 4}{3} = 3$$



PCA- Example

Step 2: Center the Data

Subtract the mean from each feature to center the data:

$$Z = X - \mu$$

The centered data is:

Sample	$Z_1 = X_1 - \mu_{X_1}$	$Z_2 = X_2 - \mu_{X_2}$
1	$1 - 2 = -1$	$2 - 3 = -1$
2	$2 - 2 = 0$	$3 - 3 = 0$
3	$3 - 2 = 1$	$4 - 3 = 1$

So, the centered data matrix Z is:

$$Z = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$



PCA- Example

Step 3: Compute the Population Covariance Matrix

The population covariance matrix is given by:

$$\text{Cov}(Z) = \frac{1}{n} Z^T Z$$

where $n = 3$ (number of samples).

1. Compute $Z^T Z$:

$$Z^T Z = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Z^T Z = \begin{bmatrix} (-1)^2 + 0^2 + 1^2 & (-1)(-1) + 0 \cdot 0 + 1 \cdot 1 \\ (-1)(-1) + 0 \cdot 0 + 1 \cdot 1 & (-1)^2 + 0^2 + 1^2 \end{bmatrix}$$

$$Z^T Z = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

2. Compute the population covariance matrix:

$$\text{Cov}(Z) = \frac{1}{3} Z^T Z = \frac{1}{3} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix}$$



PCA- Example

Step 4: Perform Eigenvalue Decomposition

We need to find the eigenvalues and eigenvectors of the covariance matrix.

1. Eigenvalues (λ):

Solve the characteristic equation:

$$\begin{aligned}\det(\text{Cov}(Z) - \lambda I) &= 0 \\ \det \left(\begin{bmatrix} \frac{2}{3} - \lambda & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} - \lambda \end{bmatrix} \right) &= 0 \\ \left(\frac{2}{3} - \lambda \right)^2 - \left(\frac{2}{3} \right)^2 &= 0 \\ \lambda^2 - \frac{4}{3}\lambda &= 0 \\ \lambda \left(\lambda - \frac{4}{3} \right) &= 0\end{aligned}$$

So, the eigenvalues are:

$$\lambda_1 = \frac{4}{3}, \quad \lambda_2 = 0$$



PCA- Example

2. **Eigenvector for $\lambda_1 = \frac{4}{3}$:**

Solve:

$$\begin{bmatrix} \frac{2}{3} - \frac{4}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} - \frac{4}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$
$$\begin{bmatrix} -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

This simplifies to:

$$-\frac{2}{3}v_1 + \frac{2}{3}v_2 = 0 \implies v_1 = v_2$$

So, the eigenvector corresponding to $\lambda_1 = \frac{4}{3}$ is:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



PCA- Example

Step 5: Form the Principal Components

The first principal component (PCA1) is the eigenvector corresponding to the largest eigenvalue

1. Normalize the Eigenvector:

The eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is **not** normalized because its magnitude is:

$$\sqrt{1^2 + 1^2} = \sqrt{2}$$

To normalize, divide each component by the magnitude:

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



PCA- Example

Step 5: Form the Principal Components

2. Project the Data onto PCA1:

Project the centered data Z onto the first principal component v_1 :

$$\text{PCA1} = Z \cdot v_1$$

$$\text{PCA1} = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{PCA1} = \begin{bmatrix} (-1) \cdot \frac{1}{\sqrt{2}} + (-1) \cdot \frac{1}{\sqrt{2}} \\ 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} \\ 1 \cdot \frac{1}{\sqrt{2}} + 1 \cdot \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{PCA1} = \begin{bmatrix} -\sqrt{2} \\ 0 \\ \sqrt{2} \end{bmatrix}$$



Explained Variance Ratio

- The Explained Variance Ratio is a key concept in PCA. It tells you the proportion of the total variance in the data that is captured by each principal component. This helps you understand how much information (variance) each principal component retains.
- Each principal component (PC) captures a certain amount of variance in the data.
- The explained variance ratio for a principal component is the fraction of the total variance that it explains. It is calculated as:

$$\text{Explained Variance Ratio} = \frac{\text{Eigenvalue of the PC}}{\text{Sum of all eigenvalues}}$$



Explained Variance Ratio

Steps to Compute Explained Variance Ratio

Step 1: Compute the Eigenvalues

- Eigenvalues (λ) are obtained from the eigenvalue decomposition of the covariance matrix.
- Each eigenvalue corresponds to a principal component and represents the amount of variance explained by that component.

Step 2: Compute the Total Variance

- The total variance is the sum of all eigenvalues:

$$\text{Total Variance} = \sum_{j=1}^p \lambda_j$$

where p is the number of features (or principal components).

Step 3: Compute the Explained Variance Ratio

- For each principal component, the explained variance ratio is:

$$\text{Explained Variance Ratio for PC}_i = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j}$$



Explained Variance Ratio

Let's say we have the following eigenvalues for a dataset with 3 principal components:

$$\lambda_1 = 5, \quad \lambda_2 = 2, \quad \lambda_3 = 1$$

Step 1: Compute the Total Variance

$$\text{Total Variance} = \lambda_1 + \lambda_2 + \lambda_3 = 5 + 2 + 1 = 8$$

Step 2: Compute the Explained Variance Ratio for Each PC

1. PC1:

$$\text{Explained Variance Ratio for PC1} = \frac{\lambda_1}{\text{Total Variance}} = \frac{5}{8} = 0.625 \text{ (or 62.5\%)}$$

2. PC2:

$$\text{Explained Variance Ratio for PC2} = \frac{\lambda_2}{\text{Total Variance}} = \frac{2}{8} = 0.25 \text{ (or 25\%)}$$

3. PC3:

$$\text{Explained Variance Ratio for PC3} = \frac{\lambda_3}{\text{Total Variance}} = \frac{1}{8} = 0.125 \text{ (or 12.5\%)}$$

In the previous example: when the eigenvector=0, PC2 explains 0% of the total variance



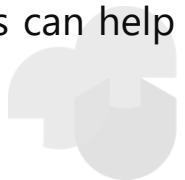
Explained Variance Ratio- Example

- PC1 explains 62.5% of the total variance.
- PC2 explains 25% of the total variance.
- PC3 explains 12.5% of the total variance.



Reconstructing Features Based on Principal Components

- **Reconstructing Features Based on Principal Components** is the process of transforming the reduced-dimensional data (projected onto principal components) back into the original feature space. This is useful for understanding how well the principal components capture the original data and for visualizing the effect of dimensionality reduction.
- **Why Reconstruct Features?**
 - ✓ **Evaluate PCA Performance:** By reconstructing the data, you can measure how much information is lost during dimensionality reduction.
 - ✓ **Visualize the Effect of PCA:** You can compare the original data with the reconstructed data to see how well the principal components approximate the original features.
 - ✓ **Data Denoising:** Reconstructing the data using only the top principal components can help remove noise.



Reconstructing Features Based on Principal Components

Steps for Reconstructing Features

Step 1: Project Data onto Principal Components

After performing PCA, the original centered data Z is projected onto the principal components to obtain the reduced-dimensional data X_{reduced} :

$$X_{\text{reduced}} = Z \cdot Q_k$$

where:

- Z = centered data matrix (mean-centered),
- Q_k = matrix of the top k eigenvectors (principal components).



Reconstructing Features Based on Principal Components

Step 2: Reconstruct the Data

To reconstruct the data in the original feature space, multiply the reduced-dimensional data X_{reduced} by the transpose of the principal component matrix Q_k :

$$Z_{\text{reconstructed}} = X_{\text{reduced}} \cdot Q_k^T$$

where:

- $Z_{\text{reconstructed}}$ = reconstructed data in the original feature space,
- Q_k^T = transpose of the principal component matrix.

Step 3: Add the Mean Back

Since the data was centered before PCA, add the mean μ back to the reconstructed data to return it to the original scale:

$$X_{\text{reconstructed}} = Z_{\text{reconstructed}} + \mu$$



Reconstructing Features Based on Principal Components- Example

Suppose we have the following centered data Z (already mean-centered):

$$Z = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Principal Components

Suppose the top principal component (PC1) is:

$$Q_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



Reconstructing Features Based on Principal Components- Example

Step 1: Project Data onto PC1

$$X_{\text{reduced}} = Z \cdot Q_1$$

$$X_{\text{reduced}} = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$X_{\text{reduced}} = \begin{bmatrix} -1 \cdot \frac{1}{\sqrt{2}} + (-1) \cdot \frac{1}{\sqrt{2}} \\ 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} \\ 1 \cdot \frac{1}{\sqrt{2}} + 1 \cdot \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 0 \\ \sqrt{2} \end{bmatrix}$$



Reconstructing Features Based on Principal Components- Example

Step 2: Reconstruct the Data

$$Z_{\text{reconstructed}} = X_{\text{reduced}} \cdot Q_1^T$$

$$Z_{\text{reconstructed}} = \begin{bmatrix} -\sqrt{2} \\ 0 \\ \sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$Z_{\text{reconstructed}} = \begin{bmatrix} -\sqrt{2} \cdot \frac{1}{\sqrt{2}} & -\sqrt{2} \cdot \frac{1}{\sqrt{2}} \\ 0 \cdot \frac{1}{\sqrt{2}} & 0 \cdot \frac{1}{\sqrt{2}} \\ \sqrt{2} \cdot \frac{1}{\sqrt{2}} & \sqrt{2} \cdot \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$



Reconstructing Features Based on Principal Components- Example

Step 3: Add the Mean Back

If the original mean was $\mu = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, add it back:

$$X_{\text{reconstructed}} = Z_{\text{reconstructed}} + \mu$$

$$X_{\text{reconstructed}} = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 2 & 3 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$



Any Question?

